Movie Recommendation System Using User Rating Data

1. Introduction

A movie recommendation system suggests movies based on user ratings using machine learning and collaborative filtering techniques.

2. Architecture Diagram / Flowchart

User Ratings Dataset → Data Preprocessing → Train Recommendation Model → Generate Recommendations → Display Recommended Movies

3. Implementation

Tools and Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Surprise Library
- Jupyter Notebook

Code

```
import pandas as pd
from surprise import Dataset, Reader, SVD
from surprise.model_selection import train_test_split
ratings=pd.read_csv('ratings.csv')
reader=Reader(rating_scale=(1,5))
data=Dataset.load_from_df(ratings[['userId','movieId','rating']],reader)
trainset,testset=train_test_split(data,test_size=0.2)
model=SVD()
model.fit(trainset)
pred=model.predict(1,10)
print(pred)
```

4. Results

Explanation of Results

The model predicts user ratings for unseen movies and recommends movies with higher predicted ratings.

Link of Runnable Project

https://colab.research.google.com/drive/1xLUFm8A0cvB0OHfb0K0WBAZj8qnas46W#scrollTo=ZQqHLQiV6Ly2

Github Link: https://github.com/ommalewar8154-wq/Movie-Recommendation-System

5. Viva Questions (One-line Answers)

Q. What is a recommendation system?
A. A system that suggests relevant items.

Q. Which algorithm is used?
A. SVD collaborative filtering.

Q. What is user rating?
A. A score given by a user to a movie.

Q. Why use collaborative filtering?
A. To recommend based on similar users.

Q. Main benefit?
A. Provides personalized movie suggestions.

Output Screenshot

```
Installing collected packages: scikit-surprise
Successfully installed scikit-surprise-1.1.5
Ratings Dataset
   userId  movieId  rating
0       1      101       5
1       1      102       4
2       1      103       3
3       2      101       4
4       2      102       5

RMSE
RMSE: 0.4007

Recommended Movie Prediction
User ID : 1
Movie ID: 104
Predicted Rating: 4.03

Top Recommended Movies
Movie 101 --> Predicted Rating = 4.32
Movie 105 --> Predicted Rating = 4.19
Movie 102 --> Predicted Rating = 4.05
Movie 104 --> Predicted Rating = 4.03
Movie 103 --> Predicted Rating = 3.81
```

Figure: Movie recommendation system output showing the ratings dataset, RMSE, predicted movie rating, and top recommended movies.